

Can You Trust Your Eyes Case Study Rubric

DS 4002 - Spring 2026 - Dev Patel

DUE: TBD

Submission format: Upload link to the GitHub repository.

Individual Assignment

General Description: Submit a link on Canvas to your GitHub repository.

Preparatory Assignments: Basic familiarity with Python and an introductory understanding of machine learning concepts.

Why am I doing this? AI-generated imagery is one of the most pressing challenges in digital media today. This case study gives you hands-on experience with a real-world computer vision problem using deep learning. You will practice the full data science workflow, from data exploration to model evaluation, and communicate your findings clearly.

What am I going to do? The GitHub repository for this case study can be found at <https://github.com/devpp101/DS4002CS3>. You will begin by downloading the image dataset from HuggingFace, which contains labeled images split into two categories: real photographs and AI-generated images. After obtaining the dataset, you will explore and preprocess the images using Python, which includes resizing images to a consistent dimension, normalizing pixel values, and splitting the data into training and testing sets.

Once your data is prepared, you will build and train a Convolutional Neural Network (CNN) using Python with either TensorFlow or PyTorch. You will train the model on your training set and tune it as needed to improve performance.

After training, you will evaluate your model's performance by computing accuracy, precision, recall, and F1-score on the test set. You will also produce a confusion matrix to visualize how well your model distinguishes between real and AI-generated images. All visualizations, metrics, and outputs should be saved to your OUTPUT folder. Your final repository should be clean, well-commented, and reproducible by anyone who clones it.

Final Deliverable Includes:

- A link to your GitHub repository, including all the data used, scripts, and output.

Tips for Success

- **Start with a simple CNN architecture before adding complexity.** Getting something working is more valuable than an elaborate model that doesn't train.

- **Start small with your data.** Training a CNN can be time-consuming. Begin with a small subset of the dataset to make sure your pipeline works before training on the full data. Once everything runs correctly, scale up.

How will I know I have succeeded? You will meet expectations on the case study when your GitHub repository follows the criteria in the rubric below:

Spec Category	Spec Details
Formatting	Your repository should include the following: • README.md file, with references • DATA Folder • SCRIPTS folder • OUTPUT folder • LICENSE.md
README.md	• <u>Goal: This file serves as an orientation to everyone who comes to your repository. It should allow them to have an idea of how the repository is organized and how they can use it.</u> • Included a section describing the software and packages used, • Included a map of your repository structure • Included step-by-step instructions to reproduce your results. • Included References if applicable.
DATA Folder	• <u>Goal: This folder contains all of the data for this project.</u> • Include the initial data • Include the final data if any cleaning has been performed on the data. • If your data does not fit in GitHub, add a README.md file explaining how to access the data, and make sure your scripts can properly output to the data folder.
SCRIPTS Folder	• <u>Goal: This folder contains all of the</u>

	<u>source code for your project.</u> • Scripts should be clearly named and commented. • Scripts should include all preprocessing steps, be able to build the CNN, and evaluate the model.
OUTPUT Folder	• <u>Goal: The folder contains all output generated by your project</u> • In a PDF document, include ALL visualizations such as plots, tables, figures, confusion matrix, and any other evaluation metrics results. ○ If using one PDF document, save each figure in order of the scripts being run. ○ If using multiple PDF documents, try to name each file based on which script is being run and what the figure entails.
LICENSE.md	• <u>Goal: This file explains the terms that they may use and cites your repository.</u> • Select a license from the GitHub options. • Usually, the MIT license is best.

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